CSE 460 Homework - 1
Exploring Astah and its Features – Submission - Bsingh55

.

Problem 1 [20 pts] - Astah Setup and Basic Class Diagram

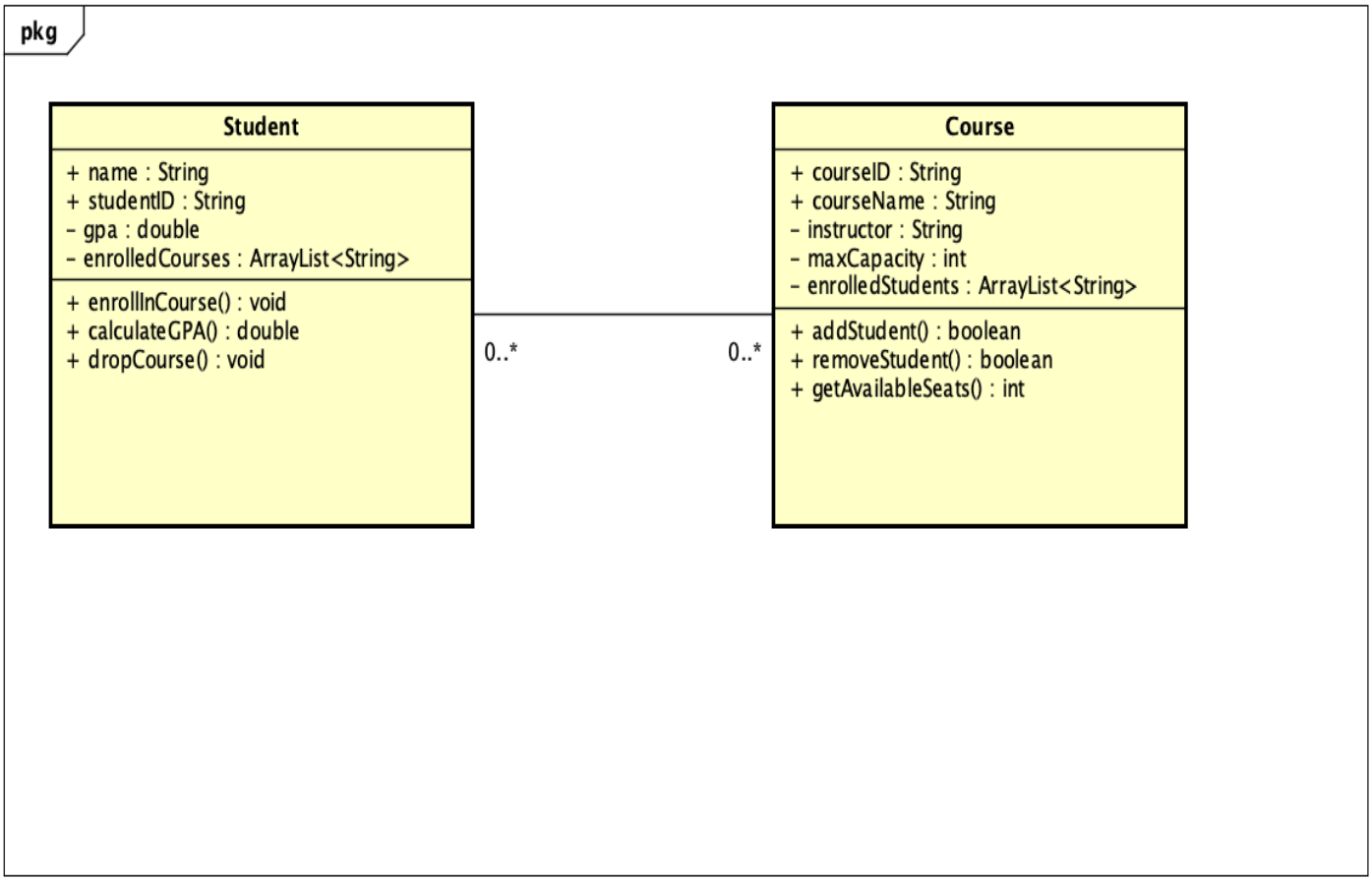

Figure 1: Basic Class Diagram showing Student–Course association.

Problem 2 [20 pts] - Advanced Class Relationships

Note: In the expanded class diagram, I intentionally removed the name attribute from the Student class because it is already defined in the abstract superclass Person with protected visibility. Since Student inherits from Person, it automatically has access to the name attribute. Removing the duplicate attribute from Student avoids redundancy and follows proper object-oriented design principles, specifically inheritance and reuse.

A.) Expand your class diagram:

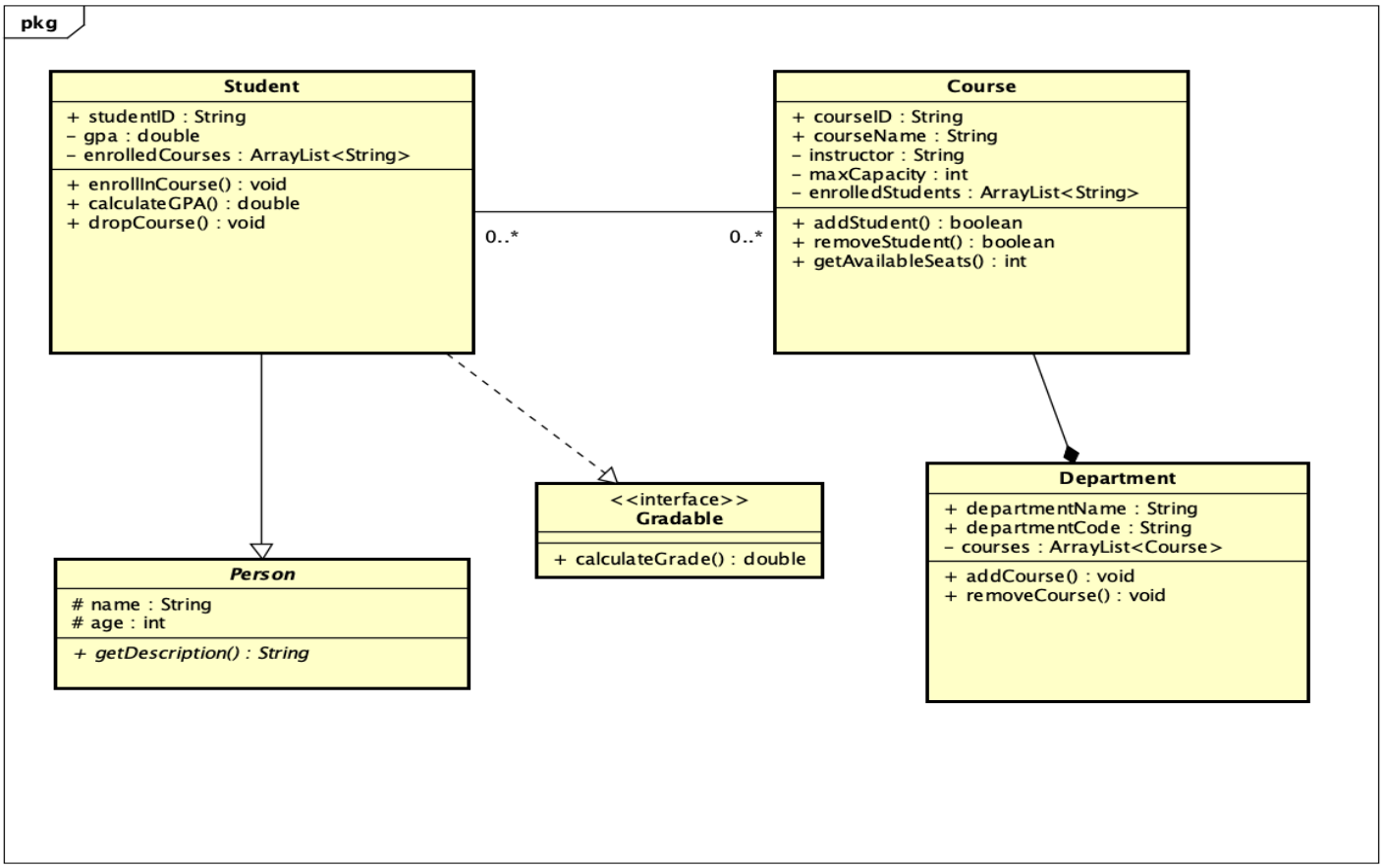

Figure 2: Expanded Class Diagram with inheritance, interface implementation, and composition.

B.) To generate Java code: [Google Drive Link (Java Code):]

Problem 3 [20 pts] - Astah Plugins

Plugin Used: PlantUML Plugin for Astah

Plugin Source: https://astah.net/product-plugins/plantuml-plugin/

1. Write your review here

I installed the PlantUML plugin, enabled it in Astah UML, and used it to generate a UML class diagram from a PlantUML description. The plugin lets me define UML elements using simple text-based syntax, which is automatically rendered into a visual UML diagram.

This plugin provides an efficient alternative to manually drawing diagrams and is especially useful for quickly prototyping designs or maintaining diagrams alongside documentation. The successful rendering of class diagrams from PlantUML syntax confirms correct installation and usage of the plugin.

2. Submit proof of your plugin usage (screenshot or exported file)

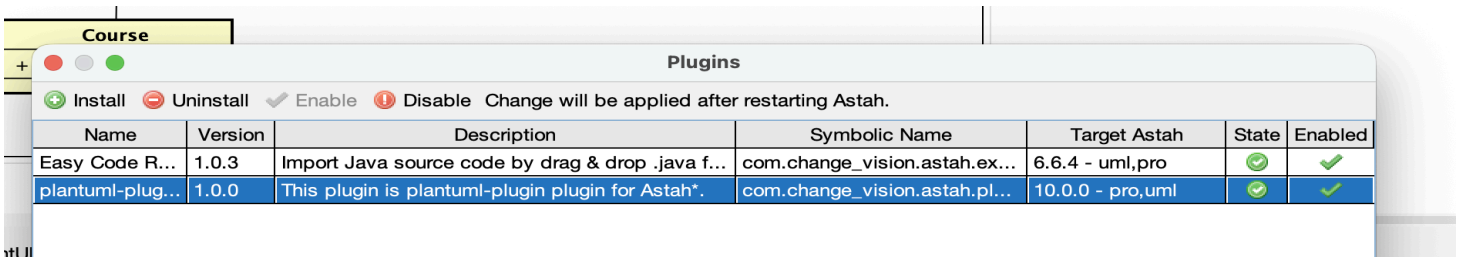

Figure 3: The PlantUML plugin installed and enabled in Astah

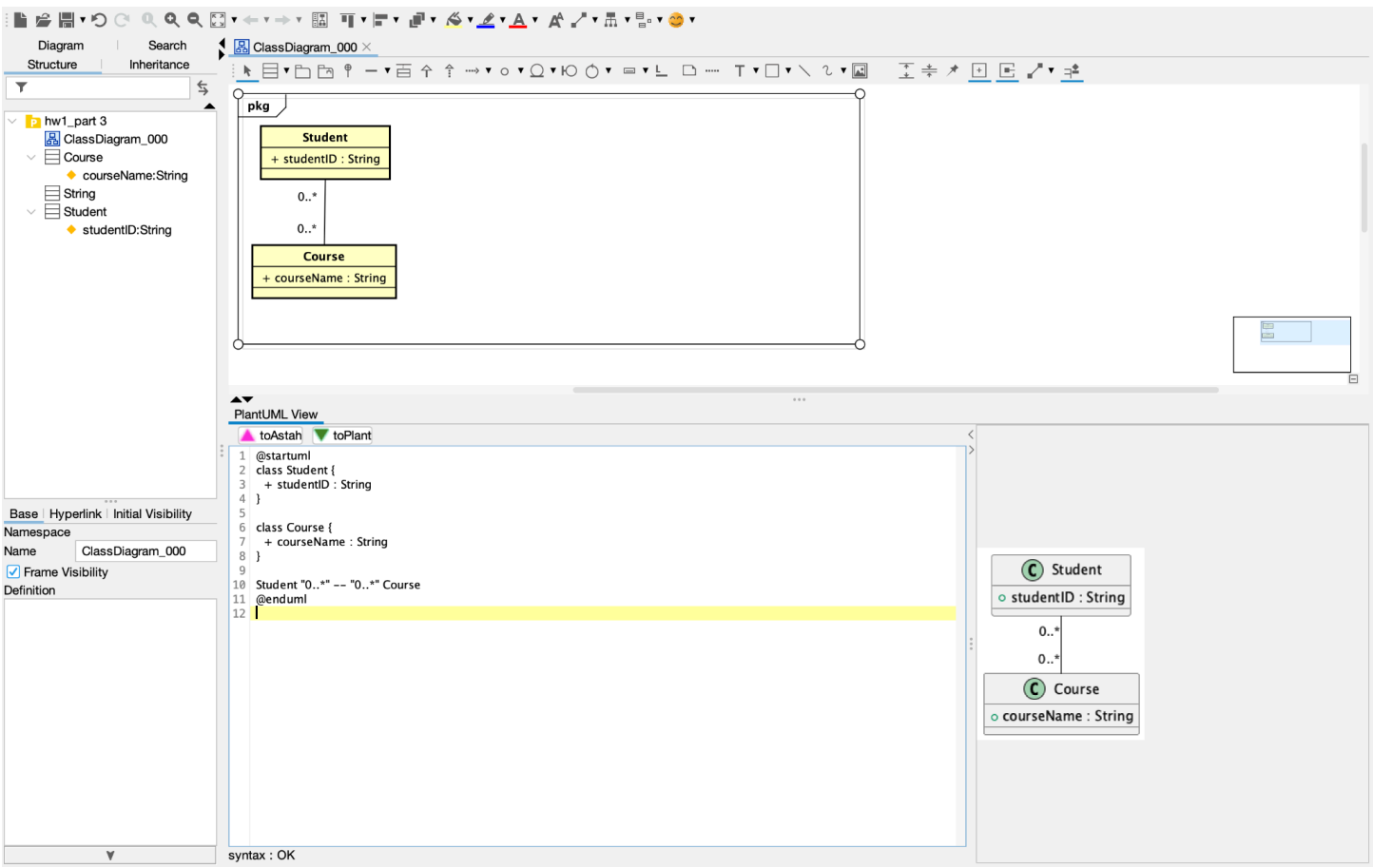

Figure 4: A UML class diagram generated from PlantUML text within Astah

Note: This diagram was intentionally kept minimal to demonstrate correct plugin installation and functionality rather than model complexity.